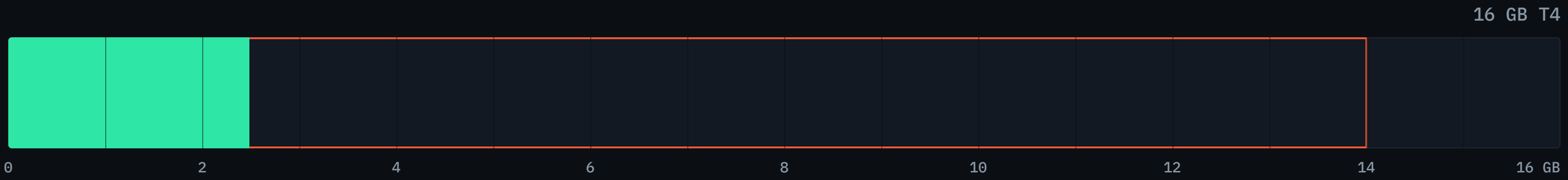
KaskyrAlmaty Soup Kernel Challenge, Decentrathon 6.0

14,345 → 2,541 MiB

Peak GPU memory of one fixed LoRA fine-tune on a free 16 GB Kaggle T4. Same training, same eval loss, 5.6× less memory, our own Triton kernels.



Change nothing about the training.
Make it smaller and faster with our own kernels.

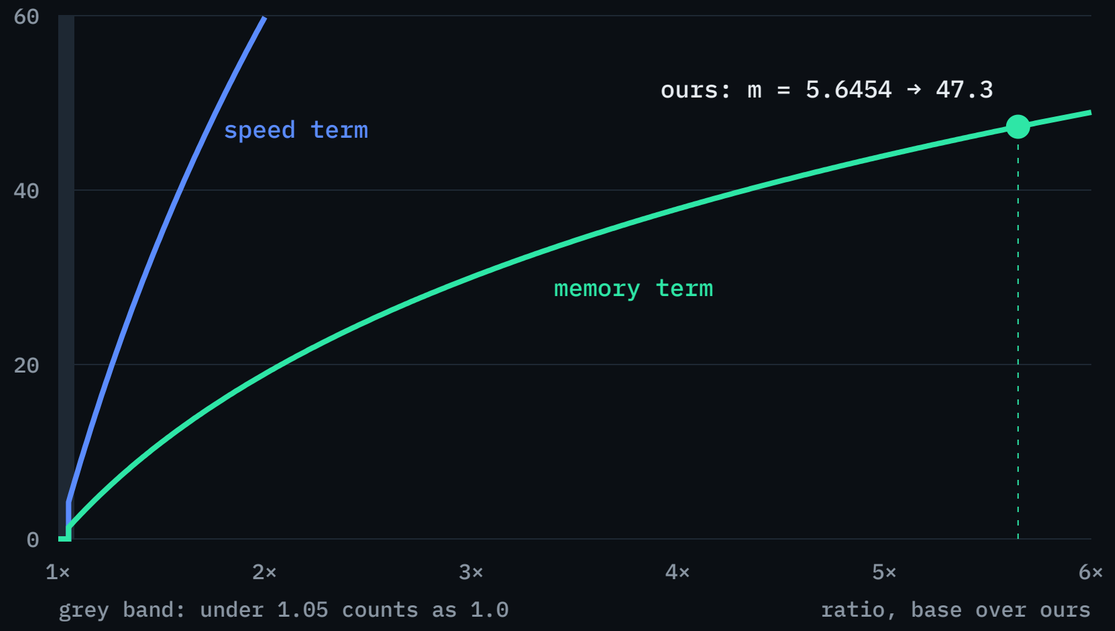
model	Qwen2-family, NF4 4-bit weights
adapter	LoRA on `q_proj` / `v_proj`
depth	`28 decoder layers`
vocabulary	`151,936 tokens`
GPU	Kaggle T4, 16 GB, free tier
frozen	☐ task, data, steps, optimizer, outputs
ours	Triton kernels, written by us

1 2 3 4 5 6 7 8 9 10

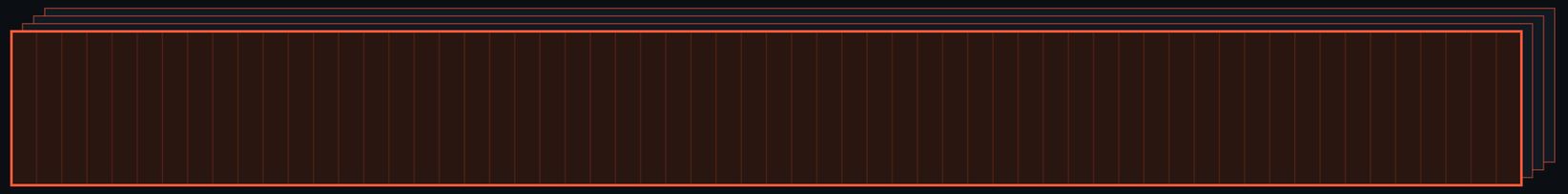
10 admission checks in `harness/check_rules.py`. Fail one and the entry scores nothing.

$$\text{Score} = 35 \cdot \ln(s')/\ln(1.5) + 30 \cdot \ln(m')/\ln(3)$$

s = speed ratio, m = memory ratio, base over ours. A ratio under 1.05 counts as 1.0.



The base run fills 14.3 of 16 GB. Copies of one tensor take 5 to 7 GB of it.



× 4 sequences
512 tokens

151,936 vocabulary columns

4 × 512 × 151,936 ≈ 311 M values
× 4 bytes (fp32) = 1.24 GB per copy

loss | its gradient | entropy | token accuracy | rewind losses

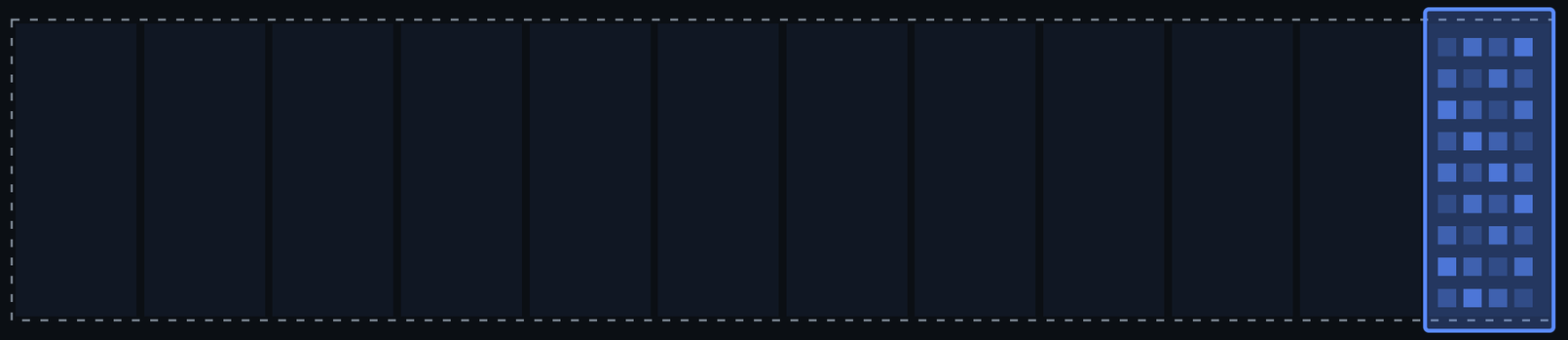
each reads or copies the full logits



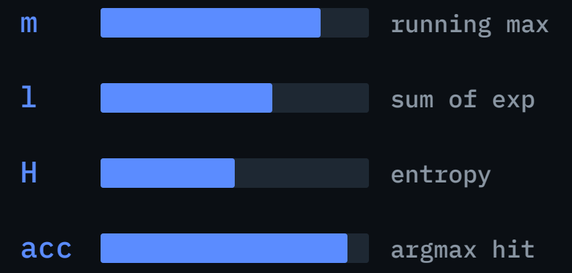
Head lane: the 151,936-wide logits never exist.

Alikhan Amirkhanov, **solution/head/**

`logits [2,048 positions × 151,936]: never allocated, 1.24 GB saved per copy`



per position: a few running values



$m' = \max(m, \max x_k)$

$l' = l \cdot e^{(m-m')} + \Sigma\, e^{(x_k-m')}$

own Triton: partial, merge, finish

Backward streams the frozen weight in slices

gather supervised positions	stream frozen weight slices	compensated hi + lo fp16 derivatives	two cuBLAS FP32 products	one cast and scatter

Checked against independent unrounded FP64 autograd, `rtol 0.01` / `atol 0.003`. Ideas from Liger fused linear CE and Apple Cut Cross-Entropy; the code is ours.

7 Triton kernels, `kernels.py`

```
head_stats_partial_kernel
head_online_merge_kernel
head_stats_finish_kernel
head_residual_max_kernel
head_residual_scale_kernel
head_ce_compensated_kernel
head_accumulate_compensated_kernel
```

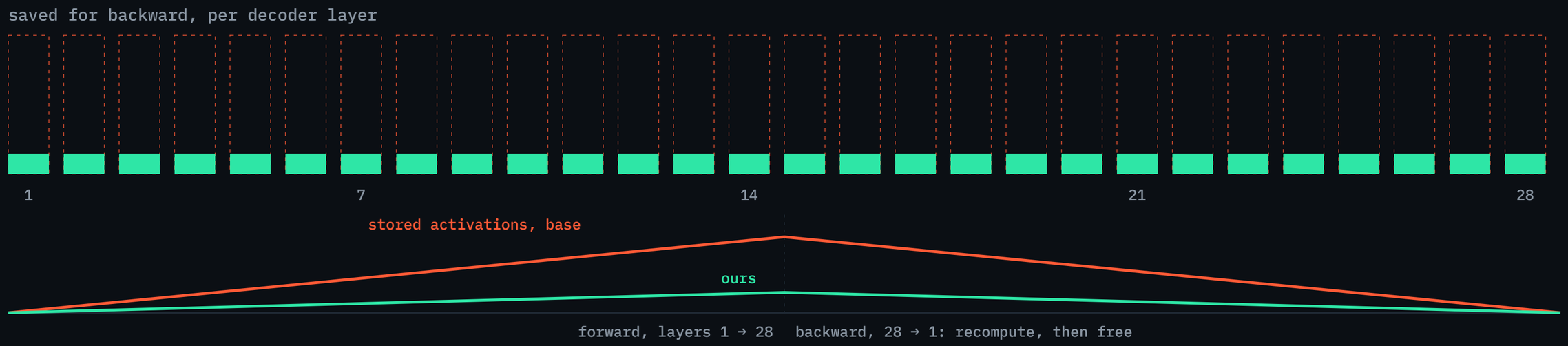
~~9,661~~

5,283 MiB

body-only solution, then with the streamed head

Body lane: forget activations in forward, rebuild them exactly in backward.

Timur Ishmurzin, `solution/body/`



Exact residual reconstruction

forward	$h = x_{fp32} + attn(x)_{fp16}$
base	stores h for backward
ours	rebuilds h from x and the actual attn branch, same ordered FP32 sum

No casting, compression or caches: the rebuilt value is the one forward computed. Own recompute of MLP, attention and norm input, with RNG replay so LoRA dropout 0.05 stays identical.

4 Triton kernels

`body_nf4_decode_weight`
`body_rmsnorm_forward`
`body_rmsnorm_backward`
`body_rope_token_kernel`

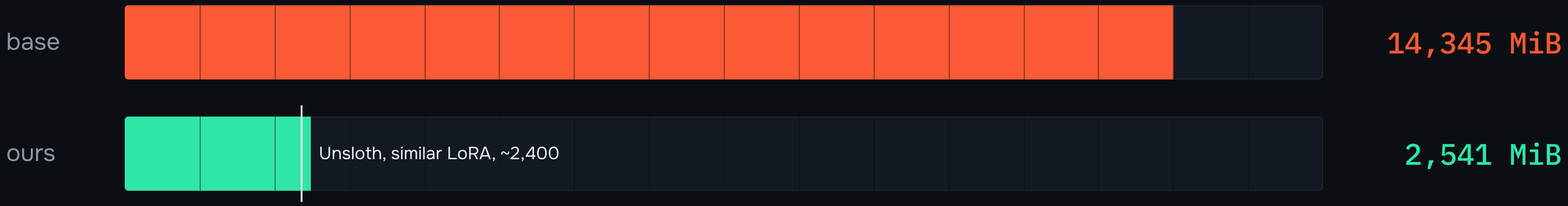
NF4 dequant for backward, RMSNorm both ways, RoPE per token.

~~5.3 GB~~

2.5 GB

replay + reconstruction on top of the streamed head

Run 52, Kaggle T4: 5.6× less memory, same loss.



5.6454

m, memory ratio. Memory term ≈ 47.3 points.

2.04964 → 2.04961

eval loss, base to ours (-0.001%)

10 of 10 admission checks; 350 solution tests and 79 Soup regression tests pass in the run

8.50%

own-kernel share of training time, 5% required

83.13 → 90.07 s

wall time of `soup train`; speed term 0

Two humans, two AI orchestrators, one measuring loop.

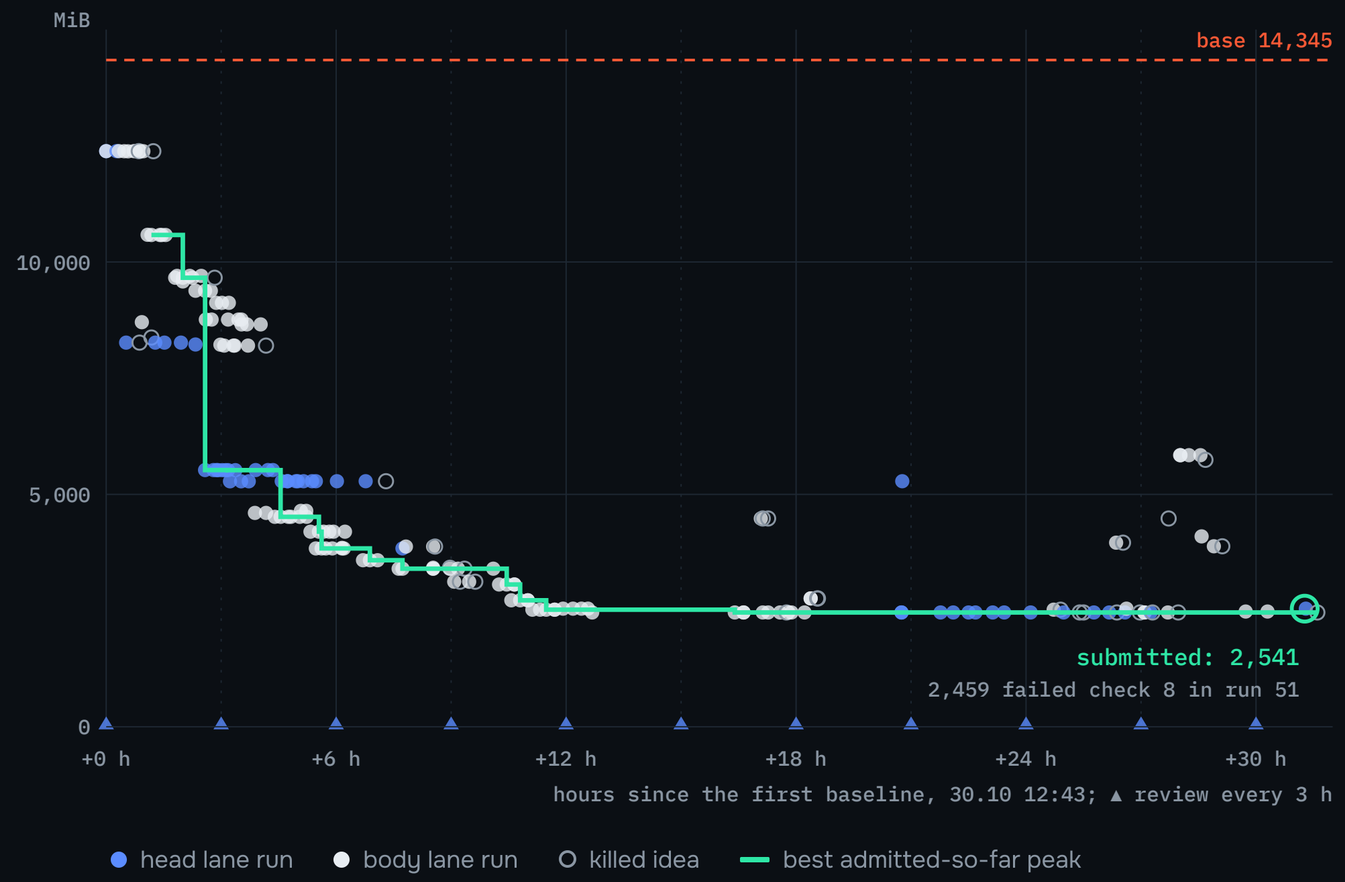
`experiment` → `Kaggle T4` → `ledger row`

~550 ledger rows across both lanes

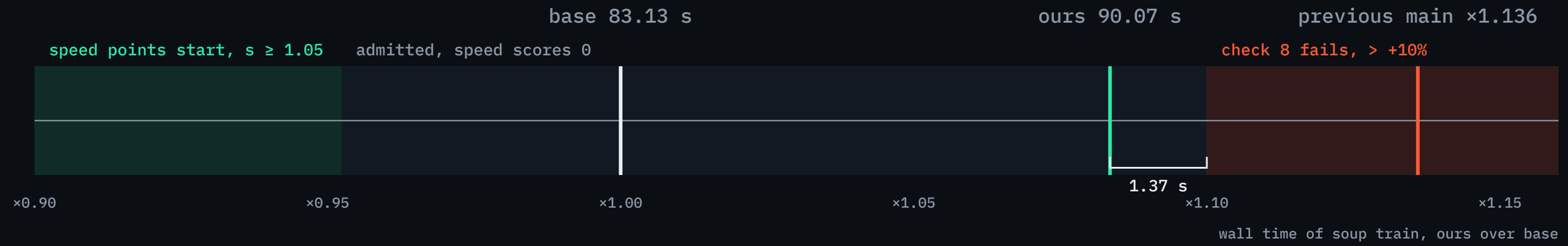
~110 experiment branches; dead ideas stay as branches with a kill row

3 h between fresh independent reviews

every merge into main gets a cross-lane review



What we did not win: speed. Recompute costs time, and the margin is thin.



Why we submit 029d932

The previous main reached 2,459 MiB and passed in run 49, then failed check 8 in run 51 at ×1.136. Commit `029d932` gives up 82 MiB and passes, 1.37 s inside the limit.

Precision is a contract

Head gradients follow an explicit FP64-reference contract, not elementwise agreement with dense autocast.

Guarded to this task

T4 fp16 path, this model. Unsupported paths fall back to the stock code.

Next: win back the 7 seconds, then go past the base.



1

Overlap or fuse the recompute

Backward rebuilds what forward dropped. Overlapping it with the GEMMs, or fusing it into them, is where the 8% went.

2

Fuse NF4 dequant into the GEMM

Today the weight is decoded to fp16, then multiplied. One kernel would skip the round trip through memory.

3

Cut start-up

About 24 s of the wall is model load and 4-bit quantization, before the first step runs.

KaskyrAlmaty

Alikhan Amirkhanov

captain, head lane: streamed vocabulary, compensated backward

Timur Ishmurzin

body lane: recompute, exact residual reconstruction, NF4, RMSNorm, RoPE

run it	kaggle.com/code/alikhanamirkhanov/kaskyr-head-1?scriptVersionId=364215273
code	github.com/BAITC-Hacks/hack-bb68f34c-kaskyralmaty
commit	029d932920765cb17f40daf651c5bf0042db74d0

14,345 → 2,541 MiB on a 16 GB T4

